Kazakh-British Technical University

Data collection and preparation

STUDENTS' INDEPENDENT STUDY (SIS) №1

Mental Health and Seasonality:
Daylight, Temperature and Well-being Across Cities

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Introduction

The aim of this project was to apply the complete data collection and preparation pipeline to a real research question: **how do seasonal factors (daylight duration, sunshine and temperature) relate to a population-level mental health indicator across cities and countries?** Data was collected from two different sources, an open API and a static web page, then cleaned, merged, analysed with descriptive statistics and correlations, and visualised with Matplotlib and Seaborn.

We tested three hypotheses:

1. Cities with a stronger seasonal swing in daylight (higher latitude) have different suicide rates than cities with stable daylight.
2. Within a city, monthly temperature follows monthly daylight (a sanity check of the weather data).
3. Sunshine hours from the API rank cities in the same order as sunshine hours from Wikipedia, and we check whether the absolute values also agree.

Tools and Data Sources

The following tools and libraries were used:

- Python and Jupyter Notebook (VS Code)
- requests: HTTP requests to the API and to Wikipedia
- BeautifulSoup: parsing HTML tables
- urllib.robotparser: checking robots.txt before scraping
- pandas and numpy: cleaning, merging and feature engineering
- scipy.stats: correlations and statistical tests
- matplotlib and seaborn: visualisations

Data sources:

1. **Open-Meteo Historical Weather API** (JSON): daily mean temperature, daylight duration and sunshine duration for 28 cities, 2019 to 2023.
2. **Wikipedia: List of countries by suicide rate** (web scraping): suicide rate per 100,000 people by country, used as the mental health indicator.
3. **Wikipedia: List of cities by sunshine duration** (web scraping): annual sunshine hours by city, used to cross-check the API.

Why suicide rate? Wikipedia has no static table of seasonal affective disorder (SAD) prevalence by country. The suicide rate is the most widely tabulated and comparable mental health outcome, so it was used as a **proxy** for population mental health. It is not a direct measure of SAD, and this is discussed in the limitations.

1. Data Collection

1.1 API: Open-Meteo

We selected 28 cities that cover a wide range of latitudes in both hemispheres, from Singapore (1.35°) and Nairobi (−1.29°) to Reykjavik (64.15°), so that the contrast in seasonal daylight is large. For each city a GET request was sent to the Open-Meteo archive endpoint with the coordinates, the period 2019-01-01 to 2023-12-31 and three daily variables: temperature_2m_mean, daylight_duration and sunshine_duration. The JSON field daily was converted into a DataFrame and a city column was added. If the API returned status 429 (rate limit), the request was repeated after a short pause. The raw result was saved to data/raw_weather.csv so that the API is called only once.

```python
def fetch_city_weather(city, lat, lon, retries=4):
    """Download daily weather for one city from the Open-Meteo archive API."""
    params = {
        "latitude": lat,
        "longitude": lon,
        "start_date": START_DATE,
        "end_date": END_DATE,
        "daily": "temperature_2m_mean,daylight_duration,sunshine_duration",
        "timezone": "auto",
    }
    for attempt in range(retries):
        resp = requests.get(API_URL, params=params, timeout=60)
        if resp.status_code == 429:          # rate limit: wait and retry
            time.sleep(5 * (attempt + 1))
            continue
        resp.raise_for_status()
        daily = resp.json()["daily"]
        out = pd.DataFrame(daily)
        out.insert(0, "city", city)
        return out
    raise RuntimeError(f"Rate limited too many times for {city}")
```

```
downloaded Oslo
downloaded Helsinki
downloaded Stockholm
downloaded Dublin
downloaded London
downloaded Berlin
downloaded Paris
downloaded Vienna
downloaded Almaty
downloaded Madrid
downloaded Rome
downloaded Athens
downloaded New York City
downloaded Toronto
downloaded Tokyo
downloaded Seoul
downloaded Cairo
downloaded Dubai
downloaded Delhi
downloaded Singapore
downloaded Nairobi
downloaded Sao Paulo
downloaded Johannesburg
downloaded Santiago
downloaded Sydney
downloaded Buenos Aires
downloaded Wellington
(51128, 5)
```

	city	time	temperature_2m_mean	daylight_duration	sunshine_duration
0	Reykjavik	2019-01-01	-0.8	15915.44	0.00
1	Reykjavik	2019-01-02	7.1	16098.18	0.00
2	Reykjavik	2019-01-03	7.7	16295.84	0.00
3	Reykjavik	2019-01-04	8.2	16507.52	0.00
4	Reykjavik	2019-01-05	4.2	16732.30	5445.38

Result: all 28 cities were downloaded successfully, **51,128 rows and 5 columns** (28 cities × 1,826 days).

1.2 Web Scraping: Wikipedia

Before scraping, robots.txt of Wikipedia was downloaded and checked with RobotFileParser. Both pages were allowed for our User-Agent. Each page was downloaded once with requests (with a descriptive User-Agent header), saved as HTML in the data folder and parsed with BeautifulSoup. Wikipedia tables often contain merged cells (rowspan and colspan) and footnote markers such as [1], so we wrote helper functions that expand merged cells into a regular grid, remove footnote markers and convert the grid into a DataFrame.

```
print("robots.txt check for the two pages we need:")
for u in ["https://en.wikipedia.org/wiki/List_of_countries_by_suicide_rate",
          "https://en.wikipedia.org/wiki/List_of_cities_by_sunshine_duration"]:
    print(" ", allowed_by_robots(u), u)
```

✓ 0.4s

```
robots.txt check for the two pages we need:
  True https://en.wikipedia.org/wiki/List_of_countries_by_suicide_rate
  True https://en.wikipedia.org/wiki/List_of_cities_by_sunshine_duration
```

Suicide rate table. All tables with class wikitable were checked, and tables with a country column and at least 40 rows were kept. Two candidate tables were found; the first one was used. Its columns are Country or territory, Region, All, Male, Female and Ratio M/F. We used the column All (both sexes).

[4] ✓ 2.0s

```
Candidate tables found: 2
Columns of the chosen table:
   Country or territory
   Region
   All
   Male
   Female
   Ratio M/F
```

	Country or territory	Region	All	Male	Female	Ratio M/F
0	World		10.3	15	6	2.5
1	Guyana	Americas	54.8	101.7	16	6.4
2	Lesotho	Africa	35	66.5	11	6.0
3	Suriname	Americas	35	62.2	12.3	5.1
4	Somalia	Africa	27.8	44.6	13.3	3.4

Sunshine table. The page contains one table per continent. Six tables with a city column and a yearly total were found and combined: **441 rows**.

```
Sunshine tables found: 6
(441, 2)
```

	city	annual_sunshine_raw
0	Gagnoa	1,823.0
1	Bouaké	2,092.0
2	Abidjan	2,296.0
3	Odienné	2,638.6
4	Ferké	2,757.0

2. Data Cleaning

Cleaning covered all required steps: renaming columns, fixing datatypes, missing values and duplicates.

- **Weather (API):** columns were renamed (time → date, temperature_2m_mean → temp_c, etc.), date was converted to datetime and numeric columns to float. There were 0 missing values and 0 duplicate (city, date) rows; short gaps of up to 3 days would be interpolated within each city. Daylight and sunshine were converted from seconds to hours, and year and month columns were added.
- **Suicide rate (scraped):** only country and rate columns were kept and renamed. The aggregate World row was removed, text values were converted to numbers, impossible values were filtered out, and country names were normalised (accents, case, aliases such as Korea, Republic of → South Korea). Result: **185 countries**, 0 missing values, 0 duplicates.
- **Sunshine (scraped):** values like 1,823.0 were converted to numbers and only plausible values (500 to 4,500 hours per year) were kept. Two duplicate city names were found (Córdoba and La Paz exist in two countries); they are not among our 28 cities. Result: **439 rows**.

```
print(weather.isna().sum())
print("Duplicate (city, date) rows:", weather.duplicated(["
```

[14] ✓ 0.0s

```
Missing values before cleaning:
city          0
date          0
temp_c        0
daylight_s    0
sunshine_s    0
dtype: int64
Duplicate (city, date) rows: 0
```

```
Missing values after cleaning:
city          0
date          0
temp_c        0
daylight_h    0
sunshine_h    0
year          0
month         0
dtype: int64
city                  object
date          datetime64[ns]
temp_c               float64
daylight_h           float64
sunshine_h           float64
year                   int16
month                   int8
dtype: object
```

	city	date	temp_c	daylight_h	sunshine_h	year	month
0	Almaty	2019-01-01	-4.8	9.036958	5.993281	2019	1
1	Almaty	2019-01-02	-2.1	9.050444	0.136906	2019	1
2	Almaty	2019-01-03	-2.0	9.065211	7.669872	2019	1
3	Almaty	2019-01-04	-1.5	9.081214	8.000000	2019	1
4	Almaty	2019-01-05	-5.2	9.098417	6.187567	2019	1

```
print(suicide.dtypes)
suicide.head()
```

✓ 0.0s

```
Missing rates: 0
Duplicate countries: 0
(185, 3)
country          object
suicide_rate    float64
country_key      object
dtype: object
```

	country	suicide_rate	country_key
0	Guyana	54.8	guyana
1	Lesotho	35.0	lesotho
2	Suriname	35.0	suriname
3	Somalia	27.8	somalia
4	Gabon	26.3	gabon

✓ 0.0s

```
Missing annual values: 0
Duplicate cities     : 2
      city annual_sunshine_raw  annual_sunshine_wiki
Córdoba                 2,905                2905.0
Córdoba                2533.9                2533.9
 La Paz                 3,148                3148.0
 La Paz                2288.9                2288.9
(439, 2)
```

	city_key	annual_sunshine_wiki
0	gagnoa	1823.0
1	bouake	2092.0
2	abidjan	2296.0
3	odienne	2638.6
4	ferke	2757.0

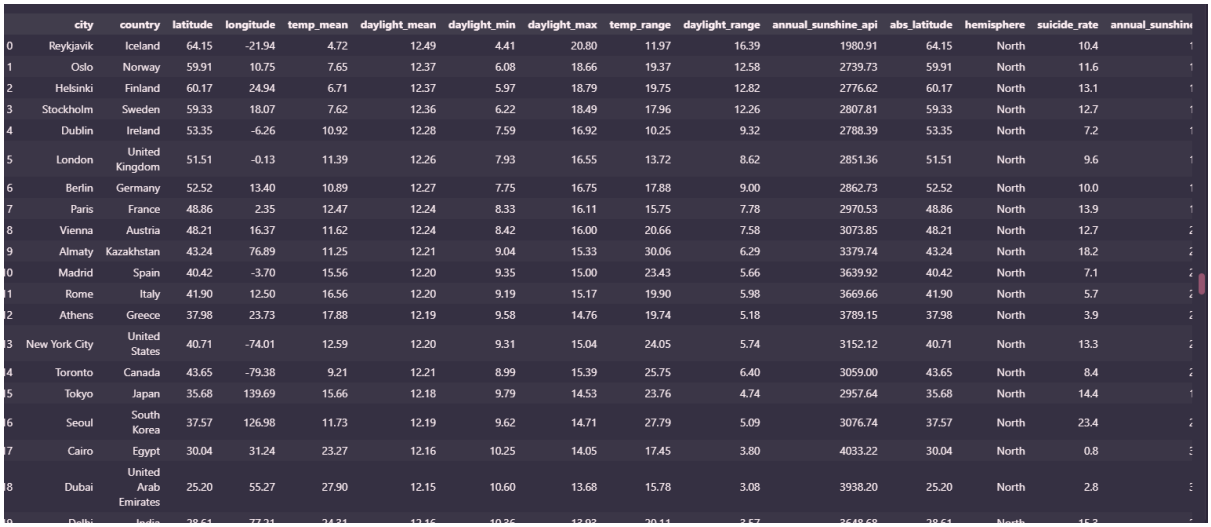
3. Merging and Feature Engineering

Daily weather was first averaged by month for each city (12 monthly means), and then every city was summarised by the following features: temp_mean, temp_range (warmest minus coldest month), daylight_mean, daylight_min (shortest month), daylight_max, **daylight_range (seasonal swing in daylight, our main seasonality measure)** and annual_sunshine_api (mean daily sunshine hours multiplied by 365). Absolute latitude and hemisphere were added.

The weather summary was joined to the suicide rate by **country** and to the Wikipedia sunshine table by **city**, using normalised name keys and a left join so that no city is lost. **All**

28 cities were matched with both a suicide rate and a Wikipedia sunshine value. The final dataset was saved as data/final_city_dataset.csv.

```
Rows: 28
Cities with a suicide rate   : 28
Cities with Wikipedia sunshine: 28
Cities without a suicide rate: []
```



	city	country	latitude	longitude	temp_mean	daylight_mean	daylight_min	daylight_max	temp_range	daylight_range	annual_sunshine_api	abs_latitude	hemisphere	suicide_rate	annual_sunshine
0	Reykjavik	Iceland	64.15	-21.94	4.72	12.49	4.41	20.80	11.97	16.39	1980.91	64.15	North	10.4	1
1	Oslo	Norway	59.91	10.75	7.65	12.37	6.08	18.66	19.37	12.58	2739.73	59.91	North	11.6	1
2	Helsinki	Finland	60.17	24.94	6.71	12.37	5.97	18.79	19.75	12.82	2776.62	60.17	North	13.1	1
3	Stockholm	Sweden	59.33	18.07	7.62	12.36	6.22	18.49	17.96	12.26	2807.81	59.33	North	12.7	1
4	Dublin	Ireland	53.35	-6.26	10.92	12.28	7.59	16.92	10.25	9.32	2788.39	53.35	North	7.2	1
5	London	United Kingdom	51.51	-0.13	11.39	12.26	7.93	16.55	13.72	8.62	2851.36	51.51	North	9.6	1
6	Berlin	Germany	52.52	13.40	10.89	12.27	7.75	16.75	17.88	9.00	2862.73	52.52	North	10.0	1
7	Paris	France	48.86	2.35	12.47	12.24	8.33	16.11	15.75	7.78	2970.53	48.86	North	13.9	1
8	Vienna	Austria	48.21	16.37	11.62	12.24	8.42	16.00	20.66	7.58	3073.85	48.21	North	12.7	2
9	Almaty	Kazakhstan	43.24	76.89	11.25	12.21	9.04	15.33	30.06	6.29	3379.74	43.24	North	18.2	2
10	Madrid	Spain	40.42	-3.70	15.56	12.20	9.35	15.00	23.43	5.66	3639.92	40.42	North	7.1	2
11	Rome	Italy	41.90	12.50	16.56	12.20	9.19	15.17	19.90	5.98	3669.66	41.90	North	5.7	2
12	Athens	Greece	37.98	23.73	17.88	12.19	9.58	14.76	19.74	5.18	3789.15	37.98	North	3.9	2
13	New York City	United States	40.71	-74.01	12.59	12.20	9.31	15.04	24.05	5.74	3152.12	40.71	North	13.3	2
14	Toronto	Canada	43.65	-79.38	9.21	12.21	8.99	15.39	25.75	6.40	3059.00	43.65	North	8.4	2
15	Tokyo	Japan	35.68	139.69	15.66	12.18	9.79	14.53	23.76	4.74	2957.64	35.68	North	14.4	1
16	Seoul	South Korea	37.57	126.98	11.73	12.19	9.62	14.71	27.79	5.09	3076.74	37.57	North	23.4	2
17	Cairo	Egypt	30.04	31.24	23.27	12.16	10.25	14.05	17.45	3.80	4033.22	30.04	North	0.8	3
18	Dubai	United Arab Emirates	25.20	55.27	27.90	12.15	10.60	13.68	15.78	3.08	3938.20	25.20	North	2.8	3

4. Exploratory Data Analysis and Results

4.1 Descriptive statistics

Variable	Mean	Min	Max
Absolute latitude (°)	39.24	1.29	64.15
Mean temperature (°C)	14.85	4.72	27.90
Seasonal daylight swing (h)	6.34	0.15	16.39
Annual sunshine, API (h)	3,272	1,981	4,033
Annual sunshine, Wikipedia (h)	2,260	1,326	3,542
Suicide rate (per 100,000)	11.14	0.80	23.40

The seasonal swing in daylight ranges from about 0.15 hours in Nairobi and Singapore to 16.39 hours in Reykjavik. Northern hemisphere cities have a larger average swing (7.24 h) than southern ones (3.63 h), because our northern sample reaches higher latitudes.

```
num_cols = ["abs_latitude", "temp_mean", "temp_range", "daylight_min",
            "daylight_max", "daylight_range", "annual_sunshine_api", "annual_sunshine_wiki",
            "suicide_rate"]

print(df[num_cols].describe().T.round(2))
print()
print("Mean values by hemisphere:")
print(df.groupby("hemisphere")[["temp_mean", "daylight_range", "suicide_rate"]].mean().round(2))
```

[19] ✓ 0.0s

```
                     count     mean     std      min      25%      50%      75%      max
abs_latitude          28.0    39.24   15.43     1.29    32.60    40.56    49.52    64.15
temp_mean             28.0    14.85    5.86     4.72    11.17    14.76    17.62    27.90
temp_range            28.0    16.37    7.22     1.67    10.84    17.66    20.25    30.06
daylight_min          28.0     9.04    1.78     4.41     8.23     9.33    10.04    12.05
daylight_max          28.0    15.38    1.98    12.19    14.25    15.02    16.22    20.80
daylight_range        28.0     6.34    3.77     0.15     4.21     5.70     7.99    16.39
annual_sunshine_api   28.0  3272.44  472.23  1980.91  2933.91  3287.59  3642.11  4033.22
annual_sunshine_wiki  28.0  2259.98  584.91  1326.20  1844.25  2066.20  2642.15  3541.80
suicide_rate          28.0    11.14    4.98     0.80     8.12    11.40    13.45    23.40

Mean values by hemisphere:
            temp_mean  daylight_range  suicide_rate
hemisphere
North           14.12            7.24         10.56
South           17.02            3.63         12.86
```

```
# Cities with the biggest and smallest seasonal swing in daylight
cols = ["city", "country", "latitude", "daylight_min", "daylight_max", "daylight_range", "temp_mean"]
print("Largest seasonal swing in daylight:")
print(df.sort_values("daylight_range", ascending=False)[cols].head(5).round(2).to_string(index=False))
print()
print("Smallest seasonal swing in daylight:")
print(df.sort_values("daylight_range")[cols].head(5).round(2).to_string(index=False))
```

✓ 0.0s

```
Largest seasonal swing in daylight:
     city country  latitude  daylight_min  daylight_max  daylight_range  temp_mean
Reykjavik Iceland     64.15          4.41         20.80           16.39       4.72
 Helsinki Finland     60.17          5.97         18.79           12.82       6.71
     Oslo  Norway     59.91          6.08         18.66           12.58       7.65
Stockholm  Sweden     59.33          6.22         18.49           12.26       7.62
   Dublin Ireland     53.35          7.59         16.92            9.32      10.92

Smallest seasonal swing in daylight:
       city              country  latitude  daylight_min  daylight_max  daylight_range  temp_mean
    Nairobi                Kenya     -1.29         12.05         12.19            0.15      18.81
  Singapore            Singapore      1.35         12.04         12.20            0.16      26.67
  Sao Paulo               Brazil    -23.55         10.71         13.55            2.85      19.26
      Dubai United Arab Emirates     25.20         10.60         13.68            3.08      27.90
Johannesburg         South Africa    -26.20         10.52         13.74            3.22      16.00
```

4.2 Hypothesis 2: the weather data is consistent (confirmed)

Within each city, monthly temperature and monthly daylight are strongly correlated (Figure 1): **mean r = 0.83** (minimum 0.54, maximum 0.93), and no city has a weak link ($r < 0.5$). The daylight swing increases almost linearly with distance from the equator (Figure 5). This confirms that the API data behaves as expected.

4.3 Hypothesis 3: API vs Wikipedia sunshine (partly confirmed)

The two sources rank cities in almost the same order: **Pearson r = 0.91 ($p < 0.001$), Spearman ρ = 0.92**. However, the absolute values do not agree. The API gives on average 3,272 hours per year and Wikipedia 2,260 hours; for individual cities the API value is **1.12 to 1.92 times** higher (1.49 on average). A paired t-test confirms that the difference is systematic ($t = 21.53$, $p < 0.001$). A likely reason is that Open-Meteo sunshine is a model estimate based on solar radiation, while Wikipedia uses weather station data, possibly for a different period.

A high correlation therefore does not mean that the values are the same: the API is good for comparing cities but not for exact hours.

```
        print("\nNot enough matched cities to compare API and Wikipedia sunshine.")
```

✓ 0.0s

```
Within city correlation of monthly daylight and monthly temperature:
  mean r = 0.832 | min r = 0.538 | max r = 0.934
  Cities with weak link (r < 0.5): []

API vs Wikipedia annual sunshine (n=28): Pearson r=0.91 (p=1.72e-11), Spearman rho=0.92
Mean annual sunshine: API=3272 h, Wikipedia=2260 h
API / Wikipedia ratio: mean=1.49, min=1.12, max=1.92
Paired t-test of the difference: t=21.53, p=1.57e-18
```

4.4 Hypothesis 1: seasonality and the suicide rate (not supported)

We computed Pearson (linear) and Spearman (rank-based) correlations between six seasonal features and the suicide rate. Because six features were tested at once and they are all strongly related to latitude, we also applied a **Bonferroni correction** (p-value multiplied by the number of tests).

Feature	Pearson r	Spearman ρ	Spearman p	Bonferroni p
Annual sunshine (API)	−0.25	−0.25	0.21	1.00
Mean temperature	−0.34	−0.24	0.22	1.00
Temperature range	0.17	0.17	0.38	1.00
Absolute latitude	−0.01	0.02	0.91	1.00
Daylight swing	0.01	0.02	0.91	1.00
Shortest daylight	−0.01	−0.02	0.91	1.00

7.3 Correlations with the mental health indicator (hypothesis 1)

We report both **Pearson** (linear) and **Spearman** (rank based, robust to outliers) correlations. With only a few dozen cities, the p-values should be read carefully.

We test six features at once, and they are strongly related to each other (all of them depend on latitude). Testing many features raises the chance of a false positive, so we also report a **Bonferroni corrected** p-value: the Spearman p-value multiplied by the number of tests (capped at 1). A feature is treated as reliably associated only if this corrected value is below 0.05.

Создать + Code + Markdown

```
features = ["abs_latitude", "daylight_range", "daylight_min",
            "temp_mean", "temp_range", "annual_sunshine_api"]

rows = []
for f in features:
    sub = df[[f, "suicide_rate"]].dropna()
    if len(sub) >= 4:
        r, p = st.pearsonr(sub[f], sub["suicide_rate"])
        rho, p_s = st.spearmanr(sub[f], sub["suicide_rate"])
        rows.append((f, len(sub), r, p, rho, p_s))

corr_table = pd.DataFrame(rows, columns=["feature", "n", "pearson_r", "pearson_p", "spearman_rho", "spearman_p"])
# Bonferroni correction for testing several features at once
corr_table["spearman_p_bonferroni"] = (corr_table["spearman_p"] * len(corr_table)).clip(upper=1)
corr_table = corr_table.sort_values("spearman_rho", key=lambda s: s.abs(), ascending=False).round(3)
corr_table
```

✓ 0.0s Python

	feature	n	pearson_r	pearson_p	spearman_rho	spearman_p	spearman_p_bonferroni
5	annual_sunshine_api	28	-0.254	0.191	-0.246	0.206	1.0
3	temp_mean	28	-0.342	0.075	-0.241	0.217	1.0
4	temp_range	28	0.168	0.392	0.172	0.382	1.0

None of the features is significantly related to the suicide rate. The closest result is mean temperature (Pearson r = −0.34, p = 0.075), which is still above 0.05. Latitude and the daylight swing show practically zero association (ρ ≈ 0.02; Figures 2 and 3). High-latitude cities (50° and above, n = 7) have a mean rate of 10.66, and the other cities (n = 21) have 11.30; the difference is not significant (Welch t-test p = 0.67, Mann-Whitney p = 0.73; Figure 4).

corr_table

0.0s

	feature	n	pearson_r	pearson_p	spearman_rho	spearman_p	spearman_p_bonferroni
5	annual_sunshine_api	28	-0.254	0.191	-0.246	0.206	1.0
3	temp_mean	28	-0.342	0.075	-0.241	0.217	1.0
4	temp_range	28	0.168	0.392	0.172	0.382	1.0
0	abs_latitude	28	-0.009	0.963	0.022	0.913	1.0
1	daylight_range	28	0.010	0.961	0.022	0.913	1.0
2	daylight_min	28	-0.009	0.962	-0.022	0.913	1.0

```
# Group comparison: high latitude cities (strong seasons) versus the rest
df["lat_group"] = np.where(df["abs_latitude"] >= 50, "High latitude (50+)", "Lower latitude (<50)")
high = df.loc[df["lat_group"].str.startswith("High"), "suicide_rate"].dropna()
low = df.loc[df["lat_group"].str.startswith("Lower"), "suicide_rate"].dropna()

print(f"High latitude: n={len(high)}, mean={high.mean():.2f}")
print(f"Lower latitude: n={len(low)}, mean={low.mean():.2f}")
if len(high) >= 3 and len(low) >= 3:
    t, p_t = st.ttest_ind(high, low, equal_var=False)
    u, p_u = st.mannwhitneyu(high, low, alternative="two-sided")
    print(f"Welch t-test: t={t:.2f}, p={p_t:.3g}")
    print(f"Mann-Whitney U: U={u:.1f}, p={p_u:.3g}")
```

0.0s

```
High latitude: n=7, mean=10.66
Lower latitude: n=21, mean=11.30
Welch t-test: t=-0.44, p=0.665
Mann-Whitney U: U=66.5, p=0.73
```

The weak negative tendency with sunshine and temperature depends on two cities: Cairo and Dubai are the sunniest cities and report the lowest rates (0.8 and 2.8); such low values may also reflect under-reporting. Without them the association disappears (Spearman $\rho = -0.06$ for sunshine and -0.07 for temperature, $p > 0.7$). The highest rates are in Seoul (23.4), Nairobi (19.1), Almaty (18.2) and Johannesburg (16.2), which have very different climates.

5. Visualizations

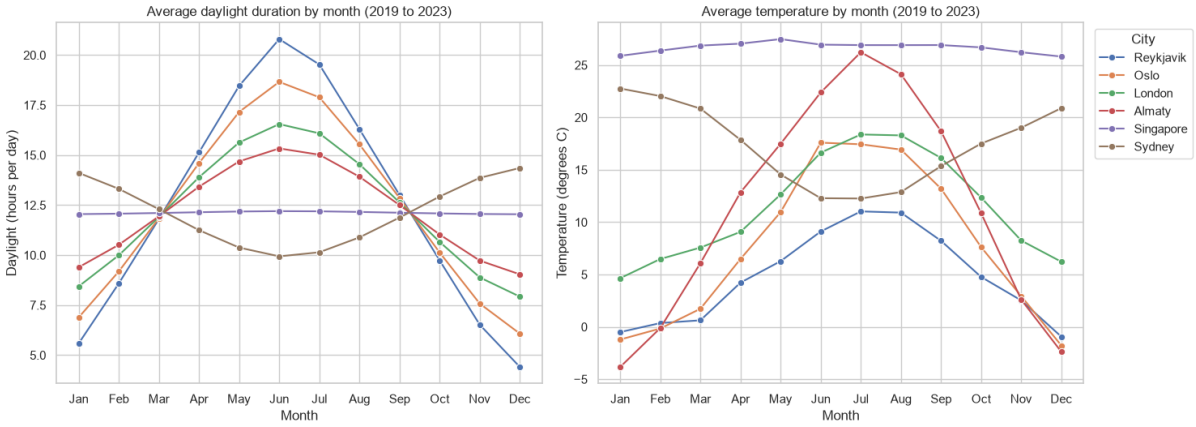


Figure 1. Seasonal profile of daylight (left) and temperature (right) for six selected cities. Daylight and temperature move together through the year, and the pattern is reversed in the southern hemisphere (Sydney).

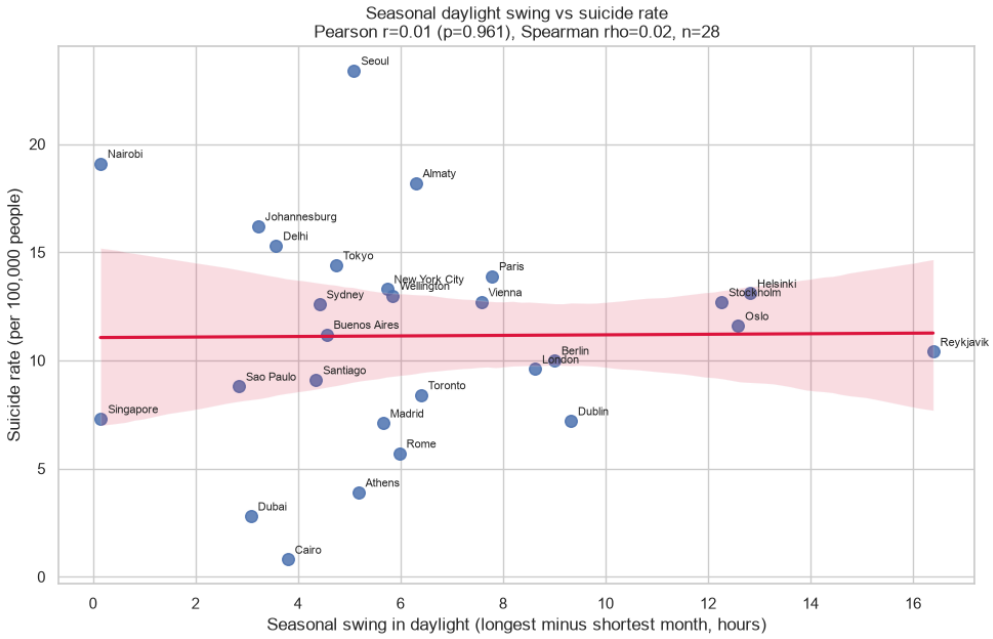


Figure 2. Seasonal swing in daylight versus suicide rate. The regression line is flat (Pearson r = 0.01, Spearman ρ = 0.02).

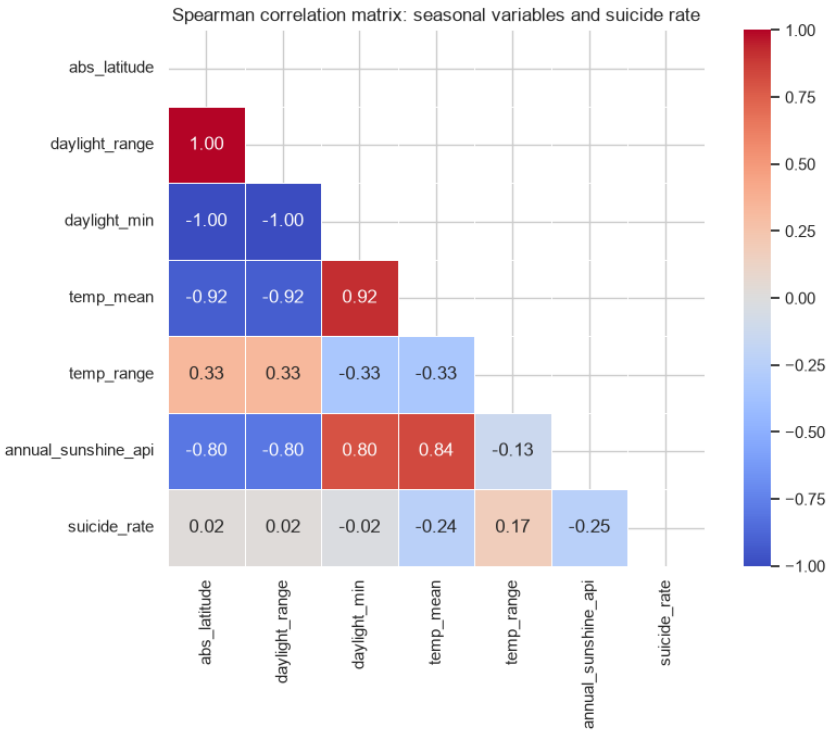


Figure 3. Spearman correlation matrix. Seasonal variables are strongly related to each other, but all correlations with the suicide rate are weak.

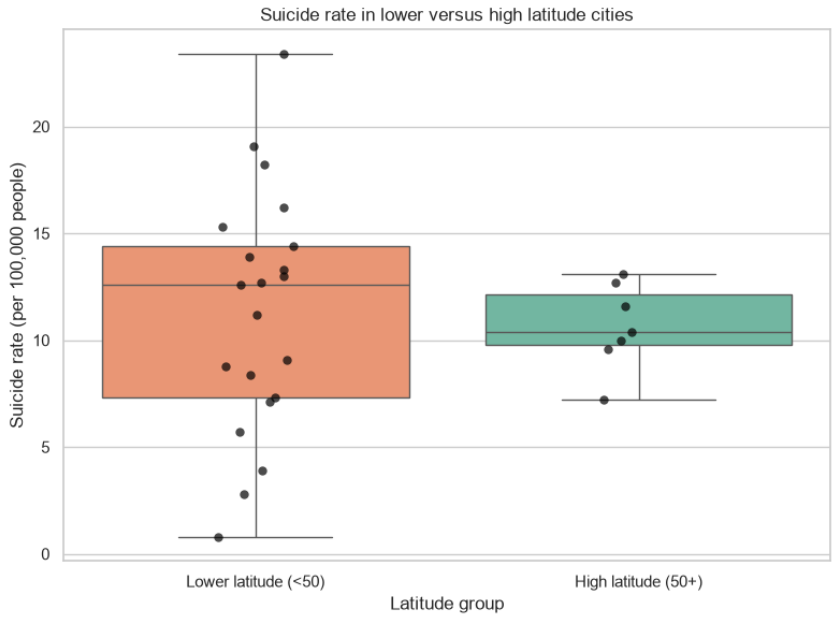


Figure 4. Suicide rate in lower-latitude and high-latitude cities. The medians are similar; lower-latitude cities show a much wider spread.

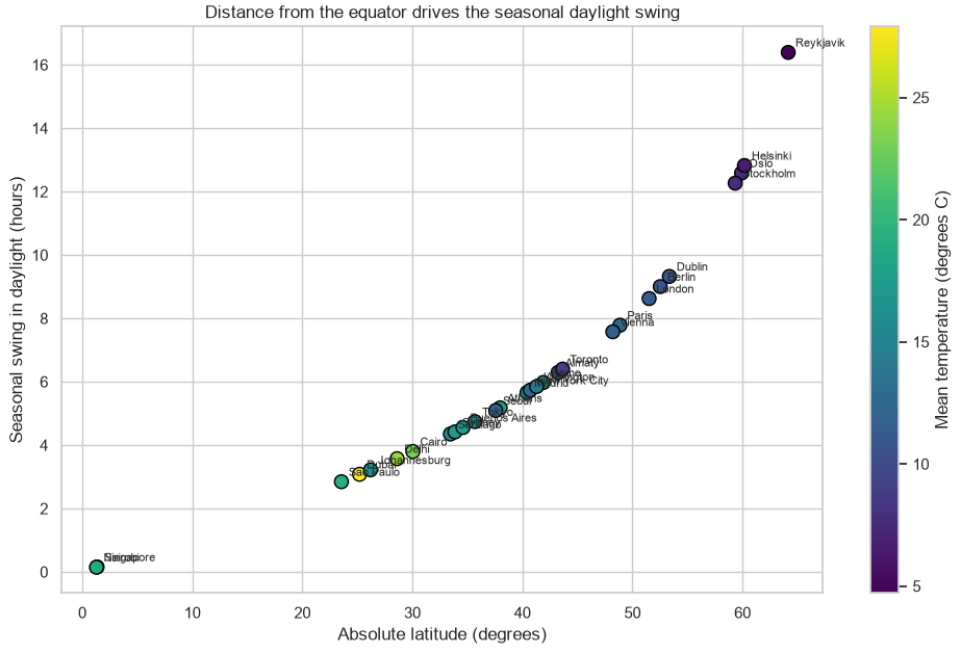


Figure 5. Absolute latitude, seasonal daylight swing and mean temperature (colour). Distance from the equator determines the daylight swing.

Key Findings

1. The pipeline collected 51,128 daily weather records from the API and two Wikipedia tables (185 countries, 439 cities), and all 28 cities were successfully merged.
2. The weather data is internally consistent: temperature follows daylight within each city (mean r = 0.83).
3. API and Wikipedia sunshine rank cities the same way (r = 0.91), but the API values are about 1.5 times higher, so correlation alone is not enough to validate a data source.

4. In our sample there is **no evidence** that daylight, seasonal swing, sunshine or temperature are related to the national suicide rate (all Bonferroni-corrected p-values are 1.00).
5. The result is sensitive to the data: an earlier run with an older version of the Wikipedia table showed a weak positive association with latitude ($\rho \approx 0.39$, $p \approx 0.04$), which disappeared after the page was updated (this earlier run is not part of the final notebook). With only 28 points, a change in the source changes the conclusion.

Limitations

- **Proxy indicator:** the suicide rate is not the same as SAD or general well-being.
- **Country vs city:** the mental health value is national, while the weather belongs to one city.
- **Small sample:** 28 cities make correlations unstable; single cities can change the result.
- **Confounding and ecological fallacy:** income, healthcare, culture, alcohol use and reporting practices differ between countries; country-level results say nothing about individuals.
- **Changing source:** Wikipedia pages are edited over time (a column name and values changed during the project), so raw HTML is cached in the data folder for reproducibility.
- **Different time frames:** the weather covers 2019 to 2023, while the Wikipedia statistics refer to the years chosen by the page authors, so the sources are not perfectly aligned in time.
- **Reporting quality:** very low rates (Cairo, Dubai) may reflect under-reporting rather than real differences.

Conclusion

In this project we built a complete data pipeline: weather data was collected from the Open-Meteo API, two tables were scraped from Wikipedia with requests and BeautifulSoup while respecting robots.txt, and the datasets were cleaned, merged by country and city, analysed and visualised. The weather data proved consistent, and the comparison of two sunshine sources showed that the API ranks cities correctly but gives systematically higher absolute sunshine hours than Wikipedia. For the main research question, we found no evidence that seasonal factors are related to the national suicide rate in our sample of 28 cities; national social and economic factors may be more important. Future work could include more cities, a city-level well-being measure and control variables such as GDP per capita from the World Bank API.

GitHub repository: [https://github.com/bohemia998/datacollection26]